FUJITSU-MONAKA: Optimized for High Performance and Energy Efficiency for AI

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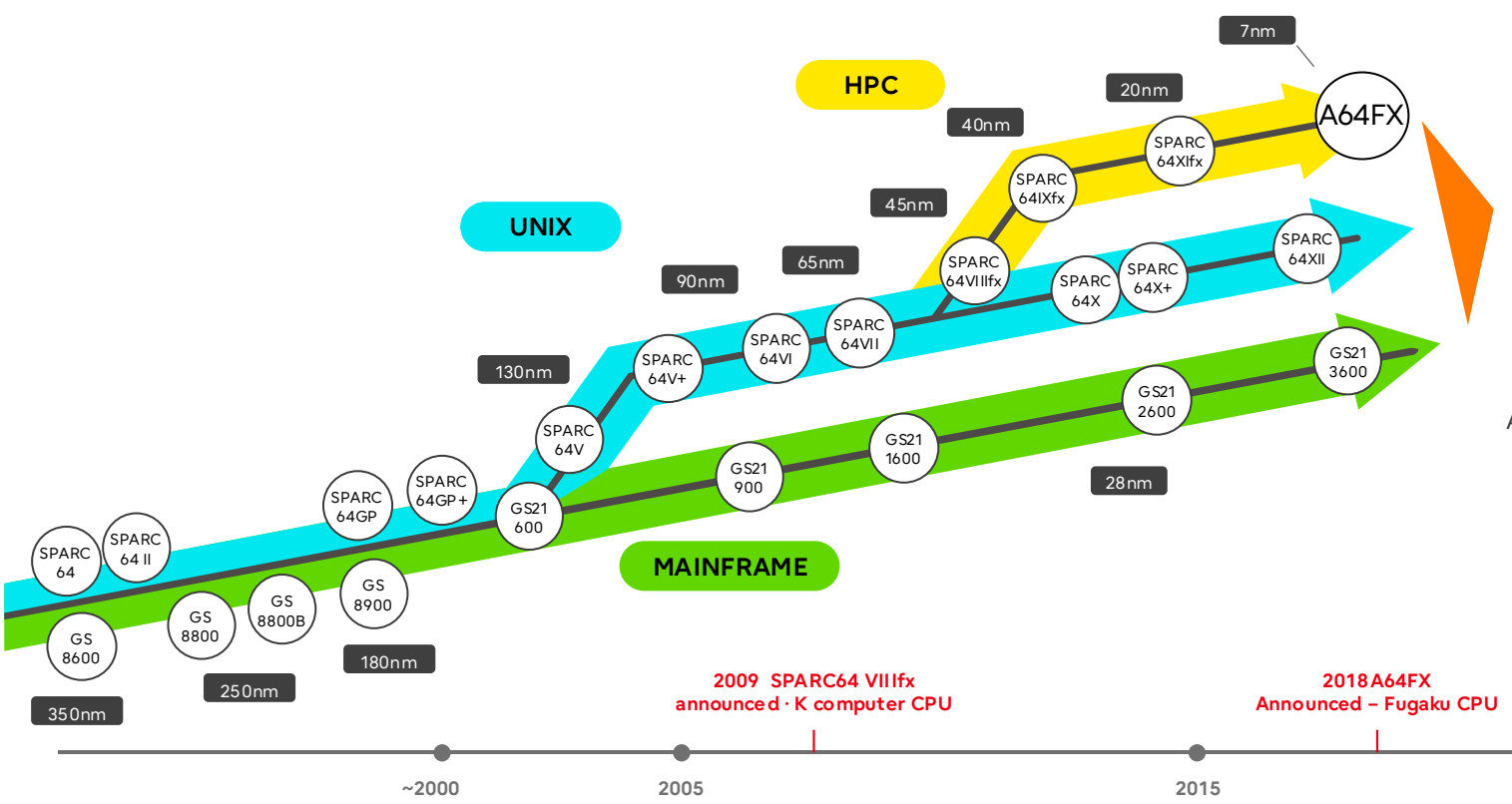
* This presentation is based on results obtained from a project subsidized by the New Energy and Industrial Technology Development Organization (NEDO).

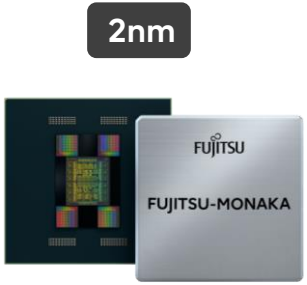


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Decades of Fujitsu Processor and Server Development

- Long-standing mainframe, UNIX server and HPC expertise shapes FUJITSU-MONAKA





FUJITSU-MONAKA

Accumulated engineering carried forward

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Power-Efficient Computing Proven at System Scale

FUJITSU



Supercomputer Fugaku

7,630,848

CPU compute cores

FUJITSU A64FX

Power-Efficient CPU with Arm SVE for high performance per watt

Massive Node system Operation

Massive scale node, processor, network, cooling, software for reliable operation

Fujitsu combines power-efficient CPU design with system-level engineering and large-scale operational expertise.

* Supercomputer Fugaku: jointly developed by RIKEN and Fujitsu

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AI/Data Center Infrastructure Demands More Than Peak Performance

FUJITSU

The adoption of generative AI is accelerating as organizations seek to strengthen competitiveness and transform operations. Beyond AI training, demand is growing for higher inference performance to maximize the value of generative AI. At the same time, new infrastructure challenges are emerging, including performance, power consumption, security and integration.

01

Performance

More inference throughput from general-purpose compute

02

Power Consumption

More compute within constrained power and cooling envelopes

03

Security

Protect sensitive data and AI workloads in use

04

Integration

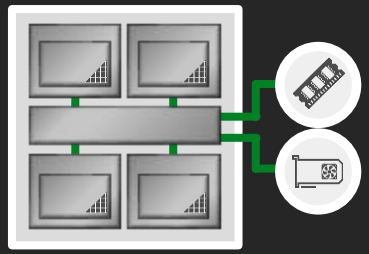
Fit into existing data-center infrastructure and software ecosystems

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FUJITSU-MONAKA Processor Overview

FUJITSU

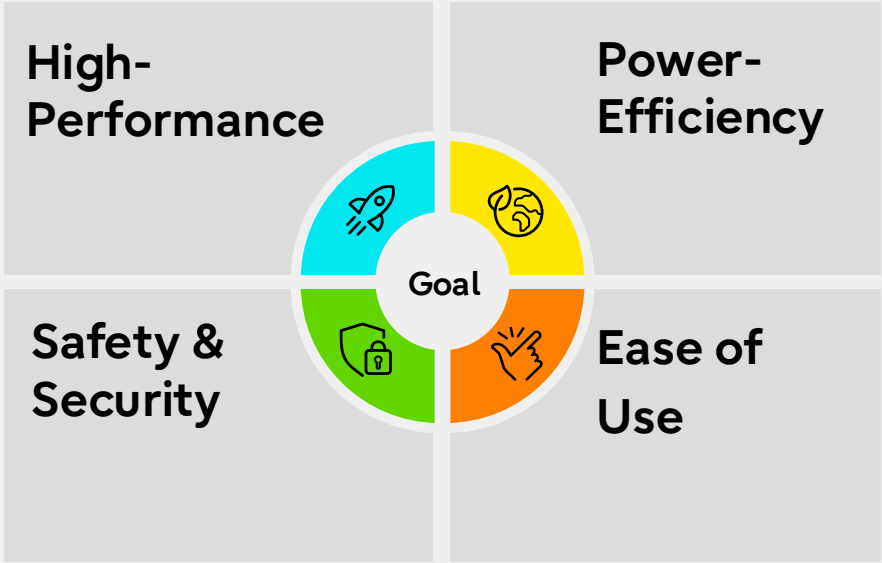
FUJITSU-MONAKA



- Arm□v9.3-A Architecture
- 3D chiplet
  - Core die 2nm
  - SRAM die/IO die 5nm
- Ultra low voltage for energy-efficiency
- DDR5 12 channels 8800MT/s
- Liquid / Air-cooling
- Arm SVE2-256bit for AI and HPC
- 144 cores x 2 sockets (288 cores per node)
- Confidential Compute Architecture for security
- PCI Express 6.0 (CXL3.0) 96 lanes

subject to change without notice

A Made-in-Japan CPU Engineered for AI Performance and Power Efficiency

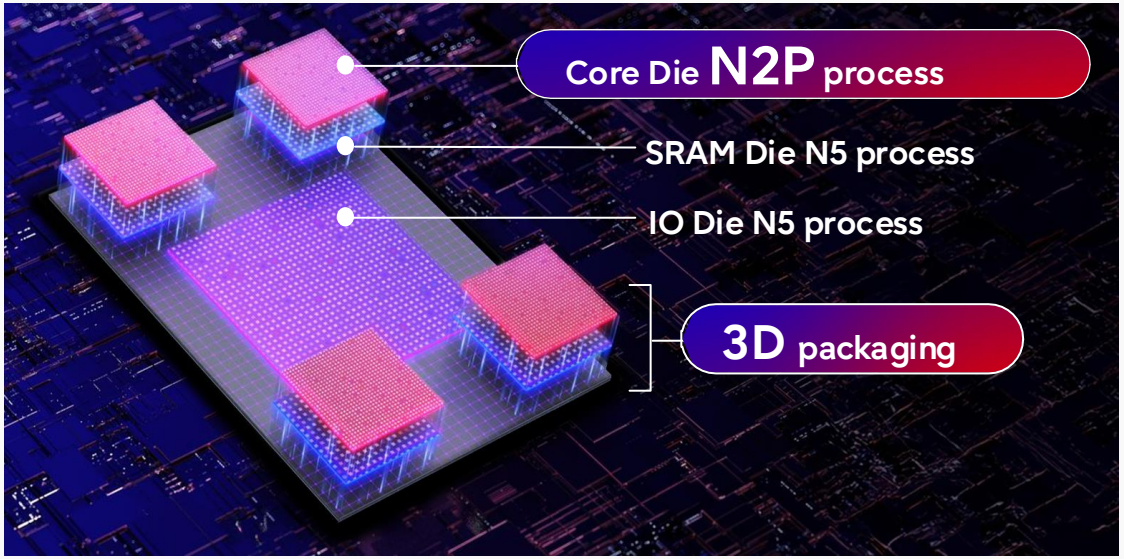


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Fujitsu's Unique Design Technologies That Create Competitiveness

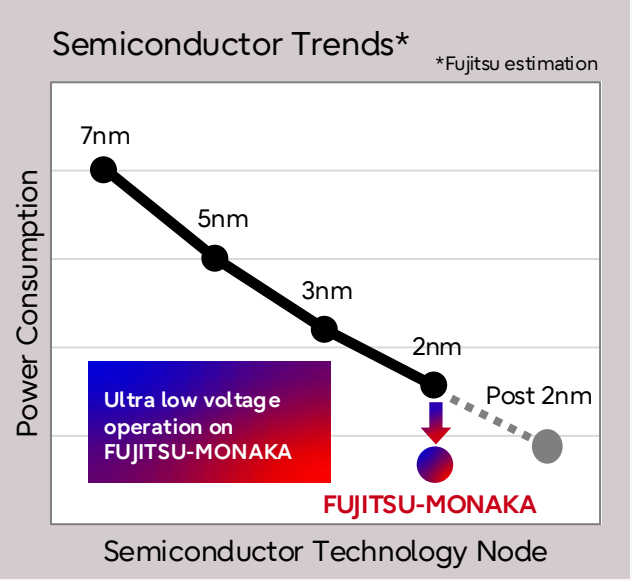
3D Many-Core Architecture

Using cutting-edge 2nm process for core die and adopting a 3D packaging design to shorten the physical data transfer distance, FUJITSU-MONAKA delivers high performance and efficiency.



Ultra Low Voltage Technology

Fujitsu's ultra low voltage technology enables energy savings beyond the 2nm generation.



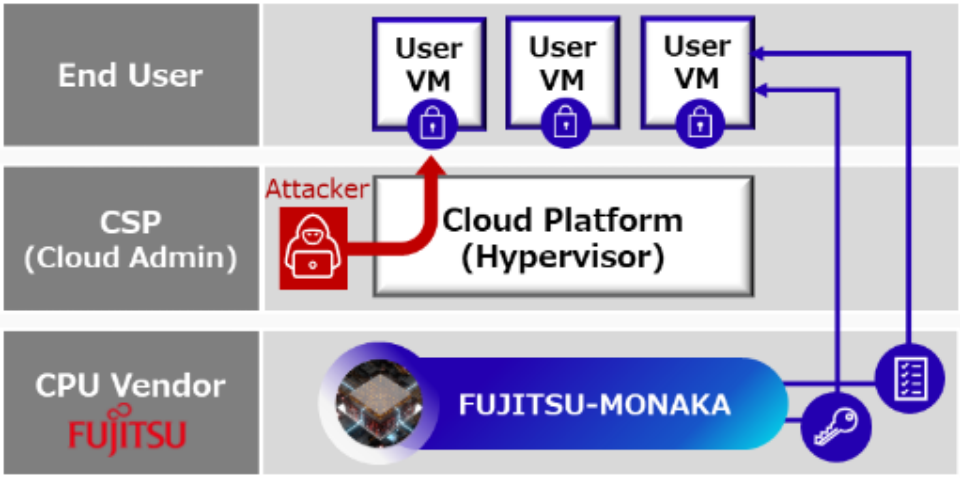
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Secure AI and Cloud Workloads by Confidential Computing at Scale

FUJITSU

4,096 User VMs

Supported with Confidential Computing Security



01 Per-VM memory encryption

A CPU-generated key protects each VM's memory.

02 CPU as the Root of Trust

Hardware and firmware anchor the trust chain.

03 Remote attestation

Platform evidence can be checked before workloads run.

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Open Standards Integration Friction and AI Inference Performance

FUJITSU

Integrate through standard interfaces, open firmware and the Arm software ecosystem.

Data Center ready TECHNOLOGY

Arm SystemReady SR

Open system firmware
- OpenBMC/EDK2 UEFI

Redfish manageability

Arm software ecosystem
- ISVs/IOVs/IFVs

Data Center ready PERFORMANCE

213.8 TOPS
INT8 peak performance
※TOPS: Tera Operations Per Second

307.0 Token/s (*)
Batch size 2
gpt-oss-20B

55.2 Token/s (*)
Batch size 2
Gemma4-31B

*Measured on a FUJITSU-MONAKA reference platform without GPU.

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Sales Begin November 2026

FUJITSU

Processor GLOBAL SALES

November 2026

Target Market : AI and Cloud data center • HPC





FUJITSU-MONAKA Reference Platform

Available now for evaluation

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Evaluate. Integrate. Build Together.

FUJITSU

01 **Evaluate FUJITSU-MONAKA**

Benchmark your workloads and facility constraints.

02 **Integrate your technology**

Connect firmware, OS, devices, memory, networking and management.

03 **Build Together**

Move from validation to deployable products and services.

Cloud • Data center • Server • ODM • Firmware • OS • Device partners

Visit Fujitsu Booth D2

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“FUJITSU-MONAKA”
Power-Efficient High-Performance CPU Development Journey

FUJITSU

Development Roadmap

2020	2027	2029
HPC, AI	Data Center, HPC, AI	Data Center, HPC, AI

7nm 

2nm 

1.4nm 

- Adopted in the supercomputer Fugaku
- First in the world to implement Arm SVE
- Low voltage operation

- Arm SVE2 support
- 3D many core
- Ultra-low voltage operation
- CCA

- SME2 for AI
- Selected for FugakuNEXT
- NVLink™ Fusion for AI/HPC

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Thank you



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